LET'S LEARN PYTHON

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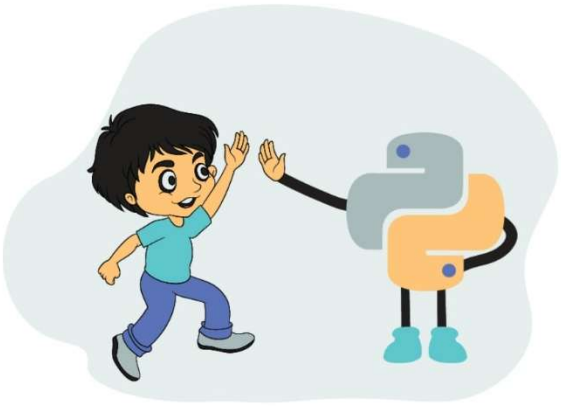
Introduction

What is Python?

Created by Guido van Rossum, Python is an object-oriented, general-purpose, high-level programming language.

The main aim of Python was to make programming easier for beginners and allow developers to get things done in fewer lines of code compared to other programming languages. Because of this simplicity, Python has become one of the most popular choices for learning programming in recent times.



Python is:

- General-purpose → can be used in web development, data science, AI, automation, and more.
- Interpreted → code runs line by line, no need for compilation.
- Interactive → you can test code directly in its shell (REPL).
- Object-Oriented → supports classes and objects for structured programming.
- Readable → uses English keywords and simple syntax rather than complex symbols.

Python is a perfect choice of programming language if you are a beginner. If you have any previous programming experience, then learning Python would be a cakewalk.

History of Python:

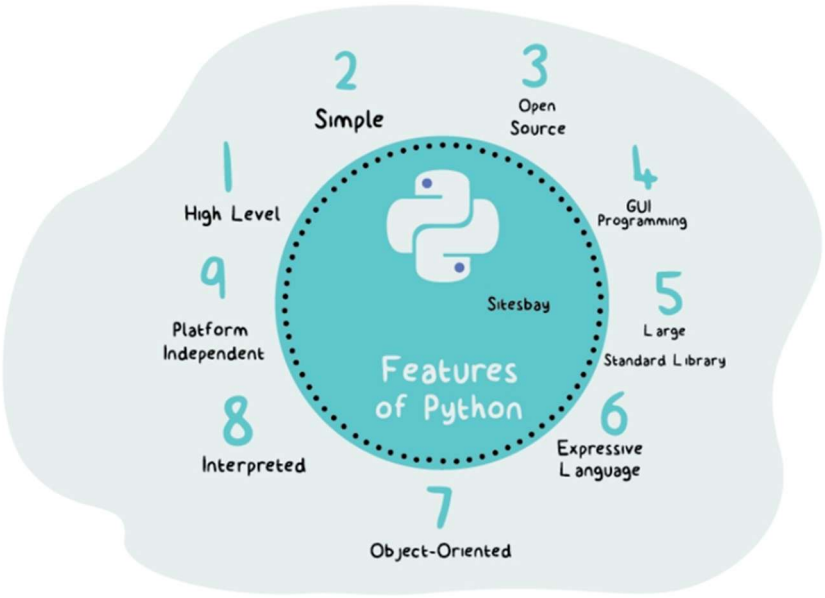
- 1985–1990: Guido van Rossum started working on Python at CWI (Centrum Wiskunde & Informatica) in the Netherlands.
- Inspiration: Python was designed as a successor to the ABC programming language. ABC had good features like exception handling and worked with the Amoeba operating system, but it also had some limitations.

- Guido wanted to keep the good parts of ABC while fixing its flaws. This led him to create a new scripting language that was simple, powerful, and practical.
- 1991: Python was officially released. The first version already included important features such as functions, exception handling, and core data types like strings, lists, and dictionaries.
- The Name "Python":
  The name did not come from the snake . Guido was a big fan of the BBC comedy show "Monty Python's Flying Circus". He wanted a name that was short, unique, and slightly mysterious, so he chose Python.

And that's how one of the world's most loved programming languages was born. Amazing, right?

Features of Python:

- Easy-to-learn - Python has few keywords, simple structure, and a clearly defined syntax. This makes it easy to catch for beginners.
- Easy-to-read - Python code is more clearly defined and visible to the eyes.
- Easy-to-maintain - Python's source code is fairly easy-to-maintain
- Broad standard library - Python's bulk libraries are very portable and cross-platform compatible with UNIX, Windows, and Macintosh.
- Interactive Mode - Python has support for an interactive mode that allows interactive testing and debugging of snippets of code.
- Portable - Python can run on a wide variety of hardware platforms and has the same interface on all of those.
- Extendable - You can add low-level modules to the Python interpreter. These modules enable programmers to add or customize their tools to be more efficient



Why Learn Python?

Do you remember that Python is a general-purpose programming language?
That means it can be used almost everywhere, and this is exactly why developers around the world love Python so much.

Python is used for:

- Web Development
- App Development
- Data Science
- Machine Learning
- Internet of Things (IoT)

...and the list goes on!

Let's deep-dive into some of these areas:

Web Development

Python is one of the top choices for backend web development (not frontend).

Django → A powerful framework used for building large, full-fledged applications.

Flask → A lightweight framework suitable for smaller applications.

Both frameworks are in high demand, and developers skilled in Django or Flask often get high-paying roles.

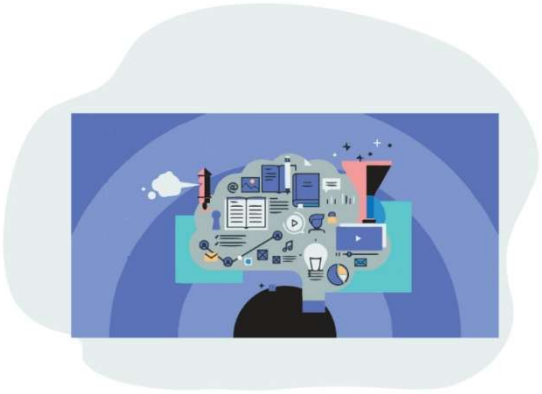
App Development

Python can also be used for mobile applications. While not the most popular choice for mobile apps, Python has a framework called Kivy, which allows you to build cross-platform apps (apps that run on both Android and iOS).

Data Science

One of the biggest reasons for Python's popularity today is its dominance in Data Science. Python offers libraries for data processing, analysis, and visualization.

It has become the go-to language for data scientists because of its simplicity + powerful tools. Skills in Python + Data Science are in very high demand in today's industry.



Machine Learning

Machine Learning might sound complicated, but Python makes it accessible even for beginners. With libraries like Keras and Scikit-learn, you can build powerful ML models without diving deep into complex math and algorithms. This makes Python a top choice for anyone who wants to step into AI and ML.

Interpreted Python & Its Versions:

Of course, we know that Python is a programming language. But did you know there are two main types of programming languages?

1. Compiled Language
2. Interpreted Language

Chances of Error

- Compiled Languages:

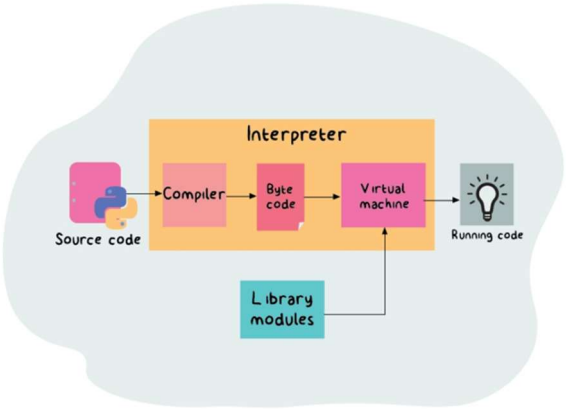
  In a compiled language, your program is first converted into machine code before it runs. If there are any errors in your code, the compiler won't let the program execute until all errors are fixed. This means you see all errors at once before running the program.

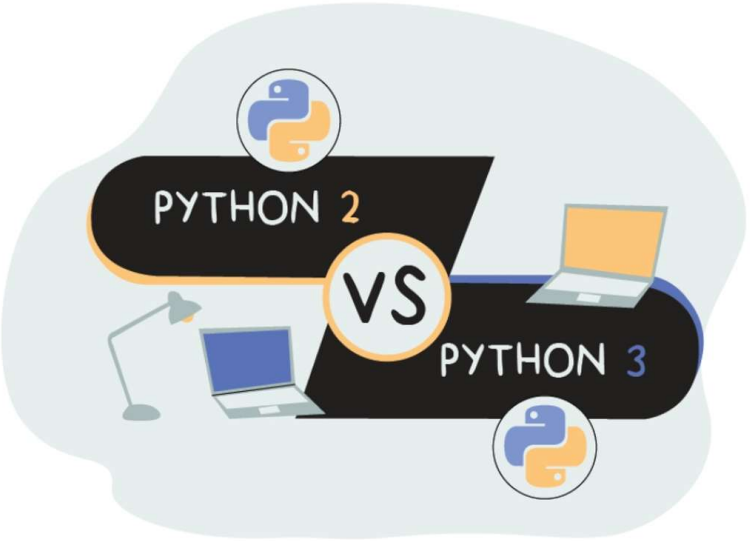
- Interpreted Languages:

  An interpreted language starts running the code immediately, even if there are errors in parts of the program. When an error is encountered, the program stops execution, and an error message shows the problem. This allows you to test code line by line.

Python is an Interpreted Language

- Most Python implementations execute instructions directly without compiling the entire program into machine code.
- The interpreter translates each statement into subroutines and then into machine code as the program runs.
- This makes Python flexible, interactive, and beginner-friendly, because you can write and test code quickly.





Python Versions

- Python 2 → Legacy Version
  - Python 2 was popular for over a decade, and some companies still use it.
  - However, it is slowly becoming obsolete.
- Python 3 → Current and Future Version
  - Python 3 fixes many of Python 2's limitations and is actively supported.
  - For beginners, it is recommended to learn Python 3, as most modern development uses it.

Installing Python & Setting up IDEs

Installing Python:

Before you start coding, you need to have Python installed on your computer. Follow these steps:

Step 1: Check if python is installed

1. Open **Command Prompt (Windows)** or **Terminal (Mac/Linux)**.
2. Type:

```bash
python --version
```

3. You should see the installed Python version
4. If not then follow the below steps

Step 2: Download Python

1. Go to the official website: https://www.python.org/downloads/
2. Click **Download Python 3.x** (latest stable version).

Step 3: Install Python

1. Run the downloaded installer.
2. **Important:** Check the box **"Add Python to PATH"** before clicking **Install Now**.
   - This makes Python accessible from the command line.
3. Follow the installation prompts.

Step 4: Verify Installation

1. Open **Command Prompt (Windows)** or **Terminal (Mac/Linux)**.
2. Type:

```bash
python --version
```

3. You should see the installed Python version, e.g., Python 3.12.0.

Python is now ready to run programs!

Installing PyCharm IDE:

PyCharm is a popular Python IDE (Integrated Development Environment) that makes coding easier for beginners and professionals.

Step 1: Download PyCharm

1. Go to https://www.jetbrains.com/pycharm/download/
2. Choose Community Edition (free) and download it.

Step 2: Install PyCharm

1. Run the installer.
2. Select installation location (default is fine).
3. Check "Add launchers to PATH" if available.

4. Finish installation.

Step 3: Launch PyCharm and Configure

1. Open PyCharm.
2. On first launch, you can choose the UI theme (light or dark).
3. Create a new project:
    - Go to File → New Project
    - Choose Python Interpreter → Select the installed Python version
    - Click Create

Writing Your First Python Program in PyCharm:

1. Right-click your project folder → **New** → **Python File**
2. Name the file hello.py
3. Type the following code:

```
1    print("Hello, World!")
```

4. Right-click inside the code editor → **Run 'hello'**
5. You should see the output:

```
Hello, World!

Process finished with exit code 0
```

Congratulations! You have successfully set up Python and PyCharm and run your first program.

Writing and Running Python Programs

Once Python is installed, you can start writing and running programs in multiple ways — no IDE required.

Using Python Interactive Shell (REPL):

Python comes with an **interactive mode**, also called the **REPL** (Read-Evaluate-Print Loop). It's great for experimenting and testing small pieces of code.

How to open Python REPL:

- Windows: Open Command Prompt → type python → press Enter
- Mac/Linux: Open Terminal → type python3 → press Enter

You should see something like:

```
Python 3.x.x (default, ... )
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

Example:

```
>>> print("Hello, Python!")
Hello, Python!
>>> 5 + 7
12
```

The >>> is called the **Python prompt**, where you can type commands directly.

Using a Python Script File:

For longer programs, you usually write your code in a **Python script file** (.py) and then run it.

Steps:

1. Open any text editor (Notepad, VS Code, etc.).
2. Write your Python code, for example:

```
print("Hello, Python!")
x = 5
y = 10
print("Sum =", x + y)
```

3. Save the file with a **.py extension**, e.g., program.py.
4. Run the program:
   - **Windows:** Open Command Prompt → navigate to the file's folder → type python program.py → Enter
   - **Mac/Linux:** Open Terminal → navigate to the folder → type python3 program.py → Enter

You should see output like:

```
Hello, Python!
Sum = 15
```

Key Points for Beginners

- Python programs can be written and run without an IDE.
- Interactive Shell is ideal for small tests and learning concepts.
- Python script files are used for larger programs and permanent code.
- Always save your files with .py extension.

Lesson Program: First program in python.

Source Code:

greetings.py

```
userName = input("Enter your name: ")
userAge = int(input("Enter your age: "))
userHeight = float(input("Enter your height in meters: "))

print("\nSummary:")
print("Name:", userName)
print("Age:", userAge, "years")
print("Height:", userHeight, "meters")
```

PYTHON BASICS

Python Syntax & Indentation

What is Syntax?

Every programming language has a set of rules that define how you should write code so that the computer can understand it. These rules are called the syntax of the language.

In Python, syntax is designed to be simple, clean, and readable, which is why beginners find it very easy to learn. Python code looks almost like plain English!

Example:

```
print("Hello, World!")
```

This is a valid Python statement. It prints a message to the screen. Unlike some other languages (like C, C++ or Java), Python doesn't need semicolons (;) at the end of statements — though you can use them if you want to put multiple statements on one line.

Statements and Lines:

Each line of code in Python is usually one statement.

Example:

```
x = 10
y = 20
print(x + y)
```

However, you can write multiple statements on a single line using semicolons:

```
x = 10; y = 20; print(x + y)
```

But this is not recommended for readability.

Indentation in Python:

In most programming languages, blocks of code (like inside an if, for, or function) are grouped using curly braces {}. But Python does it differently! Python uses indentation (spaces at the beginning of a line) to define code blocks.

For example:

```
if age >= 18:
    print("You are an adult.")
    print("You can vote.")
else:
    print("You are a minor.")
```

Here, the lines under the if and else statements are indented with 4 spaces, meaning they belong to those blocks. If you forget indentation or mix tabs and spaces, Python will show an 'IndentationError'.

Nested Indentation:

When you write blocks inside blocks, you increase the indentation level:

```
number = 10
if number > 0:
    print("Number is positive")
    if number % 2 == 0:
        print("It is also even")
    else:
        print("It is odd")
else:
    print("Number is negative")
```

Each inner block is indented more than the outer one.
This makes your code structure clear and readable.

Best Practices (PEP 8 Style Guide):

- Use 4 spaces per indentation level (no tabs).
- Be consistent throughout your code.
- Python will not tolerate inconsistent indentation!

Line Continuation:

Sometimes, your statement is too long to fit in one line. You can use a backslash (\) to continue it on the next line:

```
result = 100 + 200 + 300 + \
         400 + 500
```

You can also use parentheses ( ), brackets [ ], or braces { } for automatic continuation:

```
numbers = [
    1, 2, 3,
    4, 5, 6
]
```

Whitespace in Python:

Extra spaces around operators or inside parentheses don't affect execution.

```
value = ( 1 + 2 )
print(value)
```

But unnecessary spaces can make code look messy — so keep it neat!

Empty Blocks and the pass Statement

Python does not allow **empty code blocks**. If you need a placeholder for future code, use the pass statement.
Example:

```
if True:
    pass  # does nothing, avoids syntax error
```

Comments in Python:

What are Comments?

Comments are notes written inside a program that are ignored by the Python interpreter. They are used to explain code, make it readable, and help others (or your future self) understand what the program does.

When Python runs your code, it skips all comments completely — they are meant only for humans, not computers.

Why Use Comments?

Comments make your code self-explanatory. They are useful for:

- Describing what a block of code does
- Explaining complex logic
- Leaving reminders or "to-do" notes
- Temporarily disabling (debugging) lines of code
- Adding author or version information

Example (conceptually):

```
# This code calculates the area of a rectangle
```

Good comments help others understand your program without needing to read every single line.

Types of Comments in Python

Python supports two types of comments:

1. Single-line comments:

   A single-line comment begins with the hash symbol (#). Anything after # on that line is ignored by Python.

   Example:

   ```
   # This is a single-line comment
   print("Hello, World!")  # This is also a comment after a statement
   ```

   You can place a comment on its own line or after code on the same line.

2. Multi-line comments:

   Python doesn't have a special syntax for multi-line comments like some languages (e.g., /* ... */ in C). Instead, you can use triple quotes (''' or """) to write multiple lines of text that Python will ignore if they're not assigned to a variable or used as a docstring.

   Example:

   ```
   """ --------comments---------"""
   ''' ---------comments---------'''
   ```

Lesson Program: Comments in python.

Source Code:

comments.py

```python
# Python Program: Comments

# ---------------- SINGLE LINE COMMENTS ----------------
# This is a single-line comment
print("Hello, World!")   # You can also write a comment after code

# ---------------- MULTI-LINE COMMENTS ----------------
'''
This is a multi-line comment.
You can write as many lines as you want here.
Python will ignore these lines.
'''

"""
Another way to write multi-line comments
is to use triple double quotes.
This is also ignored by Python.
"""

# ---------------- PRACTICAL USE ----------------
# Storing user details
name = "John"   # student's name
age = 21         # student's age

print("Name:", name)
print("Age:", age)
```

Keywords and Identifiers:

What are Keywords?

Keywords are special reserved words in Python that have predefined meanings and purposes. They form the core vocabulary of the Python language and are used to define its syntax and structure.

For example, words like if, else, while, for, class, True, and False are keywords. Each of these has a special meaning and function in the Python language.

For instance:

- if → used for conditional statements
- for → used for loops
- def → used to define a function
- class → used to define a class
- True and False → represent boolean values

Python keywords are case-sensitive, so True and true are not the same — True is valid, but true will cause an error. You can view all the keywords in Python using the built-in keyword module. This helps check which words are reserved and cannot be used as variable names.

Tip: You can check all keywords by writing import keyword and then print(keyword.kwlist) in your Python shell.

What are Identifiers?

Identifiers are names used to identify variables, functions, classes, modules, or other objects in Python.

For example:

- age = 20 → here, age is an identifier.
- def greet(): → here, greet is an identifier (the name of the function).

Identifiers make it easy for you (and Python) to refer to data or functions by name.

Rules for Writing Identifiers

To ensure your names are valid, Python follows these rules for identifiers:

1. Allowed Characters:
   - Can contain letters (A–Z, a–z), digits (0–9), and underscores (_)
   - Example: name, student_1, _score
2. Cannot Start with a Digit:
   - ❌ Invalid: 2name, 123abc
   - ✅ Valid: name2, a123
3. Cannot Use Keywords:
   - You cannot name something if, class, or def because they are reserved words.
4. Case Sensitive:
   - city and City are two different identifiers.

5. No Special Characters:
   - Characters like @, #, -, $ are not allowed.
   - Example: ❌ my-name is invalid.
6. Good Practice (PEP 8):
   - Use lowercase letters for variable names (total_marks)
   - Use CapitalizedWords for class names (StudentRecord)
   - Use UPPERCASE for constants (PI = 3.14)

Tip: Checking Identifiers Python provides a simple method called .isidentifier() to check if a string is a valid identifier. Similarly, the keyword module helps to verify if a word is a Python keyword or not.

Lesson Program: Keywords and Identifiers in python.

Source Code:

keywordsAndIdentifiers.py

```
# Python Keywords and Identifiers

import keyword   # module to check Python keywords

# ---------------- KEYWORDS ----------------
print("=== Keywords in Python ===")
print("Some keywords are:", keyword.kwlist[:10])
print("Total number of keywords:", len(keyword.kwlist), "\n")

# ---------------- IDENTIFIERS ----------------
print("=== Identifiers in Python ===")

name = "Alice"
age1 = 21
_student = "John"
print("Valid identifiers ->", name, age1, _student)

city = "Mumbai"
City = "Delhi"
print("\ncity:", city)
print("City:", City)

print("\nCheck if a word is identifier or keyword:")
words = ["name", "2abc", "class", "value_1"]
for w in words:
    print(w, "-> isidentifier?", w.isidentifier(), "| iskeyword?", keyword.iskeyword(w))
```

Variables and Constants:

What is a Variable?

A variable in Python is like a container or storage box used to hold data. It stores a value that can be used and changed later in the program.

In simple terms — a variable gives a name to a value in your computer's memory so that you can reuse it easily.

Example idea:

Think of x = 10 as naming the number 10 as "x".

Characteristics of Variables

1. Dynamic Typing (No Need for Declaration):

   In Python, you don't need to declare the type of variable before using it. When you assign a value, Python automatically decides its type.

   Example idea:

   If you write age = 21, Python knows age is an integer.

   Later, you can even change it to age = "twenty-one" — Python allows that.
2. Reassignment:

   You can assign a new value to an existing variable anytime. The old value is replaced.
3. Case Sensitivity:

   Variable names are case-sensitive — meaning age, Age, and AGE are three different variables.
4. Naming Rules:
   - The name must start with a letter (A–Z or a–z) or an underscore (_).
   - It can contain letters, digits, and underscores.
   - It cannot start with a number.
   - No special characters like @, #, $, -, etc.
   - It cannot use reserved keywords (like if, while, class, etc.).
5. Good Practice in Naming:
   - Use descriptive names that tell what the data represents.

     Example: studentName is better than sn.
   - Use camalCase for variable names in Python.
6. Assigning Values:
   - You can assign a value using = operator.
     - Example idea: x = 10
   - You can assign multiple values in one line:

     x, y, z = 10, 20, 30
   - Or assign one value to multiple variables:

     a = b = c = "Python"

7. Memory Concept (Behind the Scenes):

   When you assign a variable, Python creates an object in memory and the variable name becomes a reference to that object.

Lesson Program: Variables in Python.

Source Code:

variables.py

```python
# Python Variables

# Variables store data values
name = "Alice"
age = 21
height = 5.4

print("Name:", name)
print("Age:", age)
print("Height:", height)

# Variables can be reassigned
age = 22
print("\nAfter reassignment, Age:", age)

# Valid variable name example
_valid_name = "This is valid"
print("\nValid variable name example:", _valid_name)

# Variable with underscore
my_name = "Alice Smith"
print("Variable with underscore:", my_name)

# Case-sensitive variables
city = "Mumbai"
City = "Delhi"
print("\ncity:", city)
print("City:", City)

# Multiple variables declared in one line
x, y, z = 10, 20, 30
print("\nMultiple variables -> x:", x, " y:", y, " z:", z)

# Same value to multiple variables
a = b = c = "Python"
print("Same value assigned -> a:", a, " b:", b, " c:", c)
```

What is a Constant?

A constant is a value that is not supposed to change during the execution of a program. Constants are used when a value is fixed and meaningful, such as mathematical values (PI = 3.14159) or configuration settings (MAX_USERS = 100).

In other programming languages (like C or Java), there are specific keywords like const or final to define constants but in Python, there is no built-in way to make a variable truly constant.

How Constants Work in Python?

Since Python does not have a special constant type:

- We use naming conventions to show that a variable should be treated as a constant.
- The convention is to write the constant name in all uppercase letters, often with underscores between words.

  Example:

```
PI = 3.14159
GRAVITY = 9.8
MAX_SPEED = 120
```

These are not enforced by Python, but by programmer discipline. If someone changes a constant's value later, Python will not stop them but it is considered bad practice.

Why Use Constants?

Constants make your code:

1. More readable — You instantly understand what the value represents.
2. Easier to maintain — If the value ever needs to change, you can modify it in one place.
3. Less error-prone — Prevents accidental changes to important values.

Lesson Program: Constants in Python.

Source Code:

constants.py

```python
# Python Program: Constants

# Variables
name = "Alice"
age = 21
height = 5.6

print("Variable - Name:", name)
print("Variable - Age:", age)
print("Variable - Height:", height)

# Constants (using naming convention)
PI = 3.14159
GRAVITY = 9.8
APP_NAME = "MyApp"

print("\nConstant - PI:", PI)
print("Constant - GRAVITY:", GRAVITY)
print("Constant - APP_NAME:", APP_NAME)

# Modifying constants (not recommended)
PI = 3.14
print("\n(After modifying PI) PI:", PI, "-> Not recommended!")
```

Data Types in Python:

What Are Data Types?

Every value in Python has a data type. A data type defines what kind of value a variable holds and what operations can be performed on it.

For example:

- Numbers can be added or subtracted
- Strings can be joined
- Lists can be iterated through

Python automatically decides the data type of a variable when you assign a value — you don't have to declare it explicitly.

```
x = 5          # Integer
y = "Hello"    # String
z = 3.14       # Float
```

This feature is called dynamic typing, meaning Python figures out the type during runtime.

Why Are Data Types Important?

Data types:

1. Help Python understand how to store and manage values in memory.
2. Allow the interpreter to know what operations are valid for each type.
3. Make your programs more efficient and less error-prone.

Categories of Data Types.

Python provides several built-in data types, which can be grouped into the following categories:

Category	Data Types
Numeric Types	int, float, complex
Sequence Types	str, list, tuple
Set Type	set, frozenset
Mapping Type	dict
Boolean Type	bool
None Type	NoneType

Each data type behaves differently and serves a specific purpose in Python programming.

Type Checking

You can check the data type of any variable using the built-in type() function:

```
type(10)        # <class 'int'>
type("Hello")   # <class 'str'>
```

Python also allows you to convert one data type to another (called *type casting*), which you'll learn later.

Memory and Storage

Every data type requires some amount of memory. For example, integers and floats take up more memory than Booleans or None.

You can check the memory used by a variable with the sys.getsizeof() function.

Summary

- Python determines data types automatically (dynamic typing).
- Different data types store different kinds of information.
- You can check and compare types using built-in functions.
- Knowing data types is fundamental to writing correct Python programs.

Lesson Program: Data Types in Python.

Source Code:

dataTypes.py

```python
# Python Program: DataTypes
import sys

# Basic Data Types
integerVar = 10
floatVar = 3.14
complexVar = 2 + 5j
stringVar = "Hello Python"
boolVar = True
noneVar = None
listVar = [1, 2, 3]
tupleVar = (1, 2, 3)
setVar = {1, 2, 3}
dictVar = {"one": 1, "two": 2}

print("Integer:", type(integerVar))
print("Float:", type(floatVar))
print("Complex:", type(complexVar))
print("String:", type(stringVar))
print("Boolean:", type(boolVar))
print("None:", type(noneVar))
print("List:", type(listVar))
print("Tuple:", type(tupleVar))
print("Set:", type(setVar))
print("Dictionary:", type(dictVar))

# Storage Information
print("\n----- Data Type Sizes -----")
print("Boolean:", sys.getsizeof(boolVar), "bytes")
print("Integer:", sys.getsizeof(integerVar), "bytes")
print("Float:", sys.getsizeof(floatVar), "bytes")
print("Complex:", sys.getsizeof(complexVar), "bytes")
print("String:", sys.getsizeof(stringVar), "bytes")
print("None:", sys.getsizeof(noneVar), "bytes")
```

Numeric Data Types in Python:

What Are Numeric Data Types?

Numeric data types are used to represent numbers in Python. Python provides three built-in numeric types:

1. int – Integers (whole numbers)
2. float – Floating-point numbers (decimal values)
3. complex – Complex numbers (with real and imaginary parts)

These types allow Python to perform mathematical operations and handle both small and very large values efficiently.

1. Integer (int)

- Represents whole numbers — both positive and negative.
- Example: -5, 0, 42, 1000
- There's no fixed limit on the size of an integer (unlike other languages such as C or Java).
- Python automatically converts large integers into long integers internally.
- You can perform all arithmetic operations such as addition, subtraction, multiplication, etc.

Points to remember:

- Integers don't have decimal parts.
- You can convert integers to floats or strings using float() or str().
- The function sys.getsizeof() gives the memory usage of an integer.

2. Floating-Point (float)

- Represents real numbers with a decimal point.
- Example: 3.14, 0.99, -25.6
- Internally, floats are stored in double-precision (64-bit) format.
- They can represent very small or very large numbers using scientific notation, e.g., 2.5e3 equals 2500.0.

Points to remember:

- Converting a float to an integer removes the decimal part.
- Floats can lose precision during calculations (common in all programming languages).
- sys.float_info.max gives the maximum possible float value.

3. Complex Numbers (complex)

- Used to represent numbers with a real and an imaginary part.
- Written in the form a + bj, where a is the real part and b is the imaginary part.
- Example: 3 + 4j, 2 - 5j
- You can create them directly or by using the complex() constructor.

Points to remember:

- The real and imag attributes access the real and imaginary parts respectively.

- You can perform arithmetic operations such as addition, subtraction, and multiplication on complex numbers.
- Complex numbers are very useful in mathematics, scientific computing, and signal processing.

Checking and Conversion

You can check the type of a number using:

```
type(x)
```

You can convert between numeric types using:

```
int(), float(), complex()
```

Summary

Type	Description	Example	Notes
int	Whole numbers	10, -3, 0	No decimal part
float	Decimal numbers	3.14, -0.5	Double-precision
complex	Real + imaginary	2 + 3j	Used in scientific fields

Lesson Program: Numeric Data type in Python.

Source Code:

integerDT.py

```python
# Python Program: IntegerDataType
import sys

integerNumber = 100
print("Integer value:", integerNumber)
print("Type:", type(integerNumber))
print("Size:", sys.getsizeof(integerNumber), "bytes")

bigInteger = 10**100
print("Big integer:", bigInteger)
print("Type:", type(bigInteger))
```

floatDT.py

```python
# Python Program: FloatDataType
import sys

floatNumber = 12.75
print("Float value:", floatNumber)
print("Type:", type(floatNumber))
print("Size:", sys.getsizeof(floatNumber), "bytes")
print("Max float value:", sys.float_info.max)
```

complexDT.py

```python
# Python Program: ComplexDataType
import sys

complexNumber = 3 + 4j
print("Complex value:", complexNumber)
print("Type:", type(complexNumber))
print("Size:", sys.getsizeof(complexNumber), "bytes")

complexFromInt = complex(5)
complexFromFloat = complex(2.5)
print("Complex from int:", complexFromInt)
print("Complex from float:", complexFromFloat)
```

Strings Data Type in Python:

What Is a String?

A string in Python is a sequence of characters enclosed within single quotes ' ', double quotes " ", or triple quotes ''' ''' / """ """. It can contain letters, numbers, symbols, and even spaces.

Example:

"Hello", 'Python123', """This is a string"""

In simple words — a string is used to store and manipulate text in Python.

Characteristics of Strings

1. Strings are Immutable:
   Once created, the characters inside a string cannot be changed.
   Example:

```
text = "Hello"
# text[0] = "J" ❌ -> Error
```

2. Strings Are Indexed and Ordered:
   Every character in a string has a position (index), starting from 0.
   You can access individual characters using the index -

```
text = "Python"
print(text[0])   # P
print(text[-1])  # n (last character)
```

3. Strings Can Be Multiline:
   By using triple quotes (' ' ' or " " "), you can create a string that spans multiple lines.
4. Strings Support Concatenation and Repetition:
   You can combine strings with + and repeat them using *.

```
"Hello" + " Python"   # Concatenation
"Hi! " * 3            # Repetition
```

5. Strings Can Contain Escape Sequences:
   Use a backslash (\) for special formatting -
   - \n – newline
   - \t – tab
   - \' or \" – quotes inside strings
6. Strings Support Many Built-in Methods:
   Python provides powerful methods for string manipulation –
   text.lower(), text.upper(), text.replace(), text.find(), len(text)
7. Type Casting:
   You can convert other data types (like int or float) into strings using the str() function.
8. Memory and Storage:
   Each string takes memory proportional to its length.
   Use sys.getsizeof() to check its size in bytes.

Lesson Program: String Data type in Python.

Source Code:

stringDT.py

```python
# Python Program: StringDataType
import sys

stringText = "Hello, Python!"
print("Single-line string:", stringText)

multiLineString = """This is a
multi-line string example.
It can span multiple lines."""
print("\nMulti-line string:\n", multiLineString)

userString = input("\nEnter a string: ")
print("You entered:", userString)

stringFromInt = str(123)
stringFromFloat = str(45.67)
print("\nString from int:", stringFromInt)
print("String from float:", stringFromFloat)

print("\nSize of stringText:", sys.getsizeof(stringText), "bytes")
print("Size of multiLineString:", sys.getsizeof(multiLineString), "bytes")
```

Boolean Data Type in Python:

What is a Boolean?

A Boolean in Python is a data type that represents truth values that is, whether something is True or False. It is named after George Boole, the mathematician who introduced Boolean algebra — the foundation of logical operations in computers.

Booleans are used to make decisions in programs, typically through conditions, comparisons, and control statements like if, while, etc.

Boolean Values:

Python provides two Boolean constants:

- True
- False

These are written with an uppercase first letter — lowercase versions (true, false) will cause errors.
Example:

```
x = True
y = False
```

Boolean as a Result of Expressions:

Booleans often result from comparison or logical operations:

```
5 > 3        # True
10 == 5      # False
not True     # False
```

They are the foundation of decision-making in Python.

Numeric Representation:

In Python:

- True is internally represented as 1
- False is internally represented as 0

Hence, you can even perform arithmetic with them:

```
True + True   # 2
False + True  # 1
```

Type Casting to Boolean:

Any value can be converted to a boolean using the built-in bool() function.

- bool(0), bool(0.0), bool(""), bool([]), bool(None) → False
- Any non-zero number, non-empty string, or non-empty collection → True

This is widely used for checking emptiness or existence in Python programs.

Usage in Conditions:

Boolean expressions are mainly used in conditional statements:

```
is_logged_in = True
if is_logged_in:
    print("Welcome back!")
else:
    print("Please log in.")
```

Memory and Storage

Booleans are stored as small integer objects in memory, taking very little space (usually 28 bytes in Python).

Points to Remember

- Boolean values are only `True` or `False`.
- They are case-sensitive (capital `T` and `F` are required).
- Booleans can be results of comparisons and logical operations.
- `bool()` can convert any value to True or False.
- Internally, `True = 1` and `False = 0`.

Lesson Program: Boolean Data type in Python.

Source Code:

booleanDT.py

```python
# Python Program: Boolean (bool) Data Type

import sys

# Declaration
is_active = True
is_verified = False

# Output
print("Value of is_active:", is_active)
print("Value of is_verified:", is_verified)

# Input
user_input = input("Enter True or False: ")
user_bool = user_input == "True"
print("You entered:", user_bool)

# Type Checking
print("Type of is_active:", type(is_active))
print("Type of is_verified:", type(is_verified))

# Type Casting
print("Boolean from int(0):", bool(0))
print("Boolean from int(5):", bool(5))
print("Boolean from empty string:", bool(""))
print("Boolean from 'Hello':", bool("Hello"))
print("Boolean from float(0.0):", bool(0.0))
print("Boolean from float(3.14):", bool(3.14))

# Storage
print("Memory size of is_active:", sys.getsizeof(is_active), "bytes")
print("Memory size of is_verified:", sys.getsizeof(is_verified), "bytes")
```

None Data Type in Python

What is None?

None in Python is a special constant that represents the absence of a value or a null value. It is similar to null in other programming languages (like Java, C, or JavaScript).

It means:

"This variable exists, but it doesn't have any meaningful value yet."

Type of None:

- None belongs to the NoneType data type.
- It is a singleton object, which means there is only one instance of None in an entire Python program.
- You can check this using:

```
print(type(None))  # <class 'NoneType'>
```

Usage of None:

1. Default value for variables:
   When you want to declare a variable but don't have a value yet -

```
data = None
```

2. Default function return value:
   If a function doesn't explicitly return anything, Python automatically returns None -

```
def example():
    pass
print(example())  # Outputs: None
```

3. Used in conditional checks:
   None is often used in conditions to check if something has been assigned or not -

```
if data is None:
    print("No data available")
```

4. Placeholders in data structures:
   Used to represent missing or unknown data in lists or dictionaries -

```
user = {"name": "Alice", "age": None}
```

Comparing None:

Always use is or is not when comparing with None:

```
if value is None:
    ...
```

Don't use == or != for None comparison — is checks identity, not just equality.

Memory and Storage:

None is a single constant object in memory — it takes up very little space (usually 16 bytes) and is shared everywhere in your program.

Lesson Program: None Data type in Python.

Source Code:

noneDT.py

```python
# Python Program: None Data Type

import sys

# Declaration
emptyValue = None

# Output
print("Value of emptyValue:", emptyValue)

# Type Checking
print("Type of emptyValue:", type(emptyValue))

# Comparison
if emptyValue is None:
    print("emptyValue has no value (it's None)")

# Input Example
userInput = input("Enter something (leave blank for None): ")
if userInput == "":
    userValue = None
else:
    userValue = userInput
print("You entered:", userValue, "| Type:", type(userValue))

# Storage
print("Memory size of emptyValue:", sys.getsizeof(emptyValue), "bytes")
```

Type Conversion in Python

What is Type Conversion?

Type conversion is the process of changing the data type of a value from one type to another. Python supports both automatic and manual conversions between compatible data types.

For example, converting an integer to a float or a string to an integer (when possible).

Types of Type Conversion

There are two types of conversions in Python:

1. Implicit Type Conversion (Automatic Conversion)

- Also called Type Coercion.
- Python automatically converts one data type into another without user intervention.
- This happens only between compatible data types.
- For example:
  - When an integer and a float are added, Python automatically converts the integer to float.
  - The goal is to avoid data loss and ensure smooth operations.

Example:

```
int_value = 10
float_value = 3.5
result = int_value + float_value   # int is converted to float automatically
```

Output will be a float, since float has higher precision.

2. Explicit Type Conversion (Manual Conversion)

- Also known as Type Casting.
- Done manually by the user using built-in functions.
- Common conversion functions:

Function	Description
int()	Converts data to integer (decimals are truncated).
float()	Converts data to float.
str()	Converts data to string.
bool()	Converts data to boolean (0, "", None → False).
complex()	Converts numbers to complex form.

Example:

```
num = "25"
converted = int(num)  # manually convert string to integer
```

If the conversion is not possible (e.g., "abc" → int), Python raises a ValueError.

Why Type Conversion is Important?

- To perform mathematical operations between different types.
- To format or display data properly.
- To store and process user input (since input is always a string).
- To avoid runtime errors caused by incompatible data types.

Key Points to Remember

- Implicit conversion is automatic, explicit is manual.
- Python only performs safe automatic conversions.
- Use conversion functions (int(), float(), etc.) carefully.
- Not all strings can be converted to numbers.
- Always check the type after conversion using type().

Lesson Program: Type Conversion in Python.

Source Code:

typeConversion.py

```python
# Python Program: Type Conversion

# ---------- Implicit Type Conversion ----------
print("----- Implicit Type Conversion -----")
numInt = 10       # integer value
numFloat = 3.5    # float value

# int is automatically converted to float during arithmetic operation
result = numInt + numFloat

print("Value of result:", result)
print("Type of result:", type(result))   # result is float


# ---------- Explicit Type Conversion ----------
print("\n----- Explicit Type Conversion -----")

# float to int (decimal part is removed)
floatNum = 9.99
intNum = int(floatNum)
print("Float:", floatNum, "| After int():", intNum)

# int to string
age = 21
ageStr = str(age)
print("Integer:", age, "| After str():", ageStr, "| Type:", type(ageStr))

# string to float
strNumber = "45.67"
floatNumber = float(strNumber)
print("String:", strNumber, "| After float():", floatNumber, "| Type:", type(floatNumber))
```

Input & Output in Python

In Python, input and output (I/O) refer to how a program interacts with users, taking information in (input) and showing results out (output).

Output in Python

The main function for output is print().

Basic Usage:

```
print("Hello, World!")
```

It displays text, numbers, or variables on the screen.

Multiple Values:

```
print("Sum of numbers is", 10 + 20)
```

You can print multiple values separated by commas.

Parameters in print():

1. sep → Defines the separator between printed items.

```
print("A", "B", "C", sep="-")    # Output: A-B-C
```

2. end → Defines what appears at the end of the print output (default is newline \n).

```
print("Hello", end=" ")
print("World!")    # Output: Hello World!
```

Escape Sequences:

Used to format output inside strings:

Escape	Meaning	Example Output
\n	New line	Line1 Line2
\t	Tab space	Column1 Column2
\\	Backslash	\
\' or \"	Qoutation	' or "

Formatting Output:

Python provides several ways to format text neatly:

1. String Concatenation:

```
print("Total: " + str(45))
```

2. str.format() method:

```
print("Name: {}, Age: {}".format( *args: "John", 23))
```

3. f-Strings (modern & best):

```
name = "John"; age = 23
print(f"Name: {name}, Age: {age}")
```

Input in Python

The main function for input is input().

Basic Usage:

```
name = input("Enter your name: ")
print("Hello,", name)
```

- The prompt text appears inside parentheses.
- The return value of input() is always a string, even if the user types a number.

Converting Input:

Since input() gives a string, convert it using type casting:

```
age = int(input("Enter your age: "))
height = float(input("Enter your height in meters: "))
```

Handling Invalid Input:

To prevent errors when converting invalid data:

```
try:
    num = int(input("Enter an integer: "))
except ValueError:
    print("Invalid input! Please enter numbers only.")
```

Reading Multiple Values:

You can read several values in one line:

```
a, b = input("Enter two numbers: ").split()
a = int(a); b = int(b)
```

Handling Empty Input:

If the user presses Enter without typing anything:

```
color = input("Favorite color: ").strip()
if color == "":
    color = "Not provided"
```

Input Prompt Best Practices:

Always provide clear instructions in prompts:

```
date = input("Enter date in YYYY-MM-DD format: ")
```

Lesson Program: Input & Output n in Python.

Source Code:

input&Output.py

```python
# Python Program: Input and Output

# ----- Basic print() -----
print("Hello, World!")
print("Numbers:", 10, 20, 30)    # multiple values
print()  # blank line

# ----- print() with sep and end -----
print("A", "B", "C", sep="-")    # custom separator
print("Line without newline ->", end=" ")
print("continues same line")

# ----- Basic input() -----
name = input("\nEnter your name: ")
print("Hi,", name + "!")

# ----- Type casting -----
age = int(input("Enter your age: "))      # string → int
height = float(input("Enter height in meters: "))  # string → float
print("Age:", age, "| Height:", height)

# ----- Multiple inputs -----
a, b = input("Enter two integers separated by space: ").split()
a, b = int(a), int(b)
print("Sum =", a + b)

# ----- Handling empty input -----
color = input("Favorite color (press Enter to skip): ").strip()
if color == "":
    color = "Not provided"
print("Color:", color)

# ----- Formatted output -----
item, price, qty = "Book", 120.5, 2
print(f"Using f-string -> {item}, {price}, {qty}, Total = {price * qty:.2f}")

# ----- Escape sequences -----
print("Newline ->\\n\nTab ->\\t\tExample\tTab")
```

Python Basics Summary

- Python syntax is clean and depends on **indentation** instead of braces {}. Consistent spacing (usually 4 spaces) defines code blocks like loops, functions, and conditionals.
- **Keywords** are reserved words that have special meaning in Python (like if, else, while, def, True, None). They cannot be used as variable names.
- **Identifiers** are names used for variables, functions, and classes. They must start with a letter or underscore and can't include special characters or spaces.
- **Comments** make code readable — single-line comments use #, and multi-line comments can be written using triple quotes (''' ... ''' or """ ... """).
- **Variables** store data values. Python is dynamically typed, so you don't need to declare types — for example, x = 10 or name = "John".
- **Constants** don't have built-in enforcement in Python but are written in uppercase (like PI = 3.14159) by convention to show they shouldn't be changed.
- **Data types** define the kind of data stored:
  - int for whole numbers (10)
  - float for decimals (3.14)
  - str for text ("Hello")
  - bool for logical values (True, False)
  - None represents the absence of a value or "no data."
- **Type conversion** changes one data type to another:
  - **Implicit** happens automatically (e.g., 3 + 2.5 becomes 5.5, a float).
  - **Explicit** uses functions like int(), float(), str(), etc., to manually convert types.
- **Input** is taken from the user using input(), which always returns a string — you must cast it to int or float when needed.
- **Output** is displayed using print(), which can show text and variables together. It supports parameters like sep (separator between items) and end (what to print at the end of the line).

In short, Python basics revolve around writing readable code with proper indentation, meaningful names, and correct handling of types and input/output.

Operators in Python

Operators in Python are special symbols that perform actions on data. They help Python evaluate expressions, compare values, assign data, and make logical decisions. Every operator works with one or more operands (values or variables) to produce a result.

Python categorizes operators based on the type of task they perform:

- Arithmetic operators handle mathematical tasks like addition, subtraction, multiplication, division, remainder, integer division, and exponent.
- Relational (comparison) operators compare two values and return a boolean result (True/False), useful for conditions.
- Assignment operators store values in variables and also update them using combined forms like +=, -=, *=, etc.
- Logical operators (and, or, not) combine boolean expressions and help in decision-making.
- Bitwise operators work at the binary level using AND, OR, XOR, shifts, etc.
- Membership operators (in, not in) check whether a value exists inside sequences like strings, lists, or tuples.
- Identity operators (is, is not) compare the *memory location* of two objects, not their values.

Operators are essential because they allow Python to evaluate expressions, perform calculations, make comparisons, and run logic-based programs. They appear in almost every Python script, from simple math to complex decision structures.

Lesson Program: Operators in Python.

Source Code:

operators.py

```python
# Operators in Python

print("\n===== OPERATORS IN PYTHON =====")
print("Operators perform actions on values. Examples: +, -, *, /, ==, and, or, etc.\n")

# List of operator categories
print("Types of Operators:")
print("1. Arithmetic        → +  -  *  /  %  //  **")
print("2. Relational        → ==  !=  >  <  >=  <=")
print("3. Assignment        → =  +=  -=  *=  /=  %=  //=  **=")
print("4. Logical           → and  or  not")
print("5. Bitwise           → &  |  ^  ~  <<  >>")
print("6. Membership        → in  not in")
print("7. Identity          → is  is not")

# Example variables
a, b = 10, 5
x, y = True, False
text = "Python"

print("\nExample variables:")
print("a =", a, ", b =", b)
print("x =", x, ", y =", y)
print("text =", text)

# Small demonstration
print("\nQuick examples:")
print("a + b =", a + b)                 # arithmetic
print("a > b =", a > b)                 # comparison
print("x and y =", x and y)             # logical
print("'P' in text =", 'P' in text)     # membership
print("a is b =", a is b)               # identity
```

Arithmetic Operators

Arithmetic operators are used in Python to perform mathematical calculations on numbers. These operators work with integers, floats, and even complex numbers. They are among the most frequently used tools in programming because every real-world task billing, statistics, measurements, loops, and logic uses some form of arithmetic.

The addition operator `(+)` combines two values, such as adding prices or scores. The subtraction operator `(-)` helps in finding differences—for example, remaining balance or distance left. Multiplication `(*)` is used when you need repeated addition, like calculating total cost = price × quantity. Division `(/)` always returns a decimal value, making it helpful for average calculations and ratios.

Floor division `(//)` removes the fractional part and gives only the whole number result—for example, how many full groups can be made from a total number of items. The modulus operator `(%)` returns the remainder and is widely used in tasks like checking if a number is even `(num % 2 == 0)`. Finally, the exponent operator `(**)` is used to compute powers, such as squaring a number `(n**2)` or calculating exponential growth.

Together, these operators make it easy to build logical, mathematical, and algorithmic solutions in Python.

Lesson Program: Arithmetic Operators in Python.

Source Code:

arithmeticOperators.py

```python
# Arithmetic Operators in Python

print("\n===== ARITHMETIC OPERATORS IN PYTHON =====")

# Example values
a, b = 15, 4
print("Using a =", a, "and b =", b, "\n")

# Basic arithmetic operations
print("a + b =", a + b)      # addition
print("a - b =", a - b)      # subtraction
print("a * b =", a * b)      # multiplication
print("a / b =", a / b)      # division (float)
print("a // b =", a // b)    # floor division
print("a % b =", a % b)      # remainder
print("a ** b =", a ** b)    # exponent

# User input demonstration
num1 = int(input("\nEnter first number: "))
num2 = int(input("Enter second number: "))

print("Addition:", num1 + num2)
print("Subtraction:", num1 - num2)
print("Multiplication:", num1 * num2)
print("Division:", num1 / num2)
print("Floor Division:", num1 // num2)
print("Modulus:", num1 % num2)
print("Exponent:", num1 ** num2)
```

Relational (Comparison) Operators in Python

Relational operators are used to compare two values and decide the relationship between them. These comparisons always evaluate to a Boolean result — either True or False. They are an essential part of decision-making in programming because they help control the flow of logic, especially inside if, elif, and while statements.

The **equal to** (==) operator checks whether two values are the same, commonly used to compare user input or verify a password. The **not equal to** (!=) operator checks if two values differ—useful when validating that a value should not match something.

The **greater than** (>) and **less than** (<) operators compare numeric values, such as checking if a score crosses a threshold or if temperature drops below a safe level. Similarly, **greater than or equal to** (>=) and **less than or equal to** (<=) combine equality with comparison, often used for range checks—for example, validating if a number lies between certain limits.

These operators also work on strings using alphabetical order (lexicographical comparison). For example, "cat" > "bat" is True because 'c' comes after 'b' in the alphabet. Overall, relational operators help your program "make decisions" by comparing user inputs, numeric values, strings, or even object identities.

Lesson Program: Relational Operators in Python.

Source Code:

relationalOperators.py

```python
# Relational (Comparison) Operators in Python

print("\n===== RELATIONAL OPERATORS IN PYTHON =====")

# Example values
a, b = 10, 20
print("Using a =", a, "and b =", b, "\n")

# Basic comparisons
print("a == b ->", a == b)   # equal to
print("a != b ->", a != b)   # not equal
print("a > b  ->", a > b)    # greater than
print("a < b  ->", a < b)    # less than
print("a >= b ->", a >= b)   # greater or equal
print("a <= b ->", a <= b)   # less or equal

# User comparison example
x = int(input("\nEnter first number: "))
y = int(input("Enter second number: "))

print(x, "==", y, "->", x == y)
print(x, "!=", y, "->", x != y)
print(x, ">", y, " ->", x > y)
print(x, "<", y, " ->", x < y)
print(x, ">=", y, "->", x >= y)
print(x, "<=", y, "->", x <= y)

# String comparison demo
print("\nString comparison examples:")
print("'apple' == 'apple' ->", "apple" == "apple")
print("'apple' < 'banana' ->", "apple" < "banana")
print("'cat' > 'bat' ->", "cat" > "bat")
```

Assignment Operators in Python

Assignment operators are used to store values in variables and update those values efficiently. The simplest form is the **= operator**, which assigns a value to a variable. For example, `score = 50` stores the number 50 in `score`.

Python also provides **compound assignment operators**, which combine an operation with assignment in a shorter form. For example, writing `count += 1` means "increase count by 1," which is cleaner than writing `count = count + 1`. These operators exist for addition, subtraction, multiplication, division, modulus, exponent, and floor division.

There are also **bitwise assignment operators**, used mostly in low-level or performance-oriented tasks. They work on the binary representation of numbers. For example, `x &= 1` performs a bitwise AND with 1 and updates `x` with the result.

Assignment operators help make code shorter, more readable, and more efficient by reducing repetition—for instance, updating running totals, adjusting values in loops, or modifying flags. They work only on variables (not literal values) and update the existing variable in place.

Lesson Program: Assignment Operators in Python.

Source Code:

assignmentOperators.py

```python
# Assignment Operators in Python

print("\n===== ASSIGNMENT OPERATORS IN PYTHON =====")

# Simple assignment
a = 10
print("Initial a =", a)

# Arithmetic assignment operators
a += 5   # add and assign
a -= 3   # subtract and assign
a *= 2   # multiply and assign
a /= 4   # divide and assign
a %= 3   # modulus assign
print("After various arithmetic assignments, a =", a)

# Floor division and exponent
a = 10
a //= 3
a **= 2
print("After floor division and exponent assignments, a =", a)

# Bitwise assignment operators
a = 5
a &= 3
a |= 2
a ^= 1
a <<= 1
a >>= 2
print("Final value after bitwise assignments, a =", a)

# User input example
x = int(input("\nEnter a number: "))
x += 10
x *= 2
x //= 5
print("Final x after assignment operations =", x)
```

Logical Operators in Python

Logical operators are used to combine conditions and make decisions in programs. They always work with boolean values (True or False) and return a boolean result.

Python has three logical operators:

- and → True only if *both* conditions are True
- or → True if *at least one* condition is True
- not → Reverses the truth value (True → False, False → True)

These operators are commonly used inside if statements, comparisons, and decision-making logic. Example: checking if a number lies in a range or combining multiple checks together. They make conditions more readable and allow writing compact expressions like:

```
1       age > 18 and citizen == True
```

Logical expressions evaluate from left to right, but and has higher precedence than or, meaning it is evaluated first. Parentheses can be used to control evaluation order. These operators help the program decide what to do based on multiple conditions.

Lesson Program: Logical Operators in Python.

Source Code:

logicalOperators.py

```python
# Logical Operators in Python

print("\n===== LOGICAL OPERATORS IN PYTHON =====")

# Example values
a, b, c = 10, 20, 5
print("Using a =", a, ", b =", b, ", c =", c, "\n")

# Logical AND
print("a > b and b > c ->", (a > b) and (b > c))

# Logical OR
print("a > b or b > c  ->", (a > b) or (b > c))

# Logical NOT
print("not(a > b) ->", not(a > b))

# Combined expression
print("(a > b) or (a < c and b > c) ->", (a > b) or (a < c and b > c))

# Boolean value example
x, y = True, False
print("\nBoolean examples:")
print("x and y ->", x and y)
print("x or y  ->", x or y)
print("not x   ->", not x)

# User input example
num = int(input("\nEnter a number: "))

if num > 0 and num < 10:
    print("Number is between 1 and 9.")
elif num <= 0 or num >= 10:
    print("Number is not in the range 1-9.")

print("\nLogical operators: and / or / not\n")
```

Bitwise Operators in Python

Bitwise operators work directly on the binary form of integers. Every number is converted into bits (0s and 1s), and the operation happens on each bit. They are mostly used in low-level tasks like optimization, hardware communication, networking, encryption, etc.

Common Bitwise Operators:

- & (AND) → bit becomes 1 only if both bits are 1
- | (OR) → bit becomes 1 if any one bit is 1
- ^ (XOR) → bit becomes 1 if bits are different
- ~ (NOT) → flips bits (0→1, 1→0)
- << (Left Shift) → moves bits left (adds 0s at right)
- >> (Right Shift) → moves bits right (drops rightmost bits)

Example:

5 → 0101
3 → 0011
5 & 3 → 0001 (1)
5 | 3 → 0111 (7)
5 ^ 3 → 0110 (6)

Lesson Program: Bitwise Operators in Python.

Source Code:

bitwiseOperators.py

```python
# Bitwise Operators in Python

print("\n===== BITWISE OPERATORS IN PYTHON =====")

# Example values
a, b = 5, 3   # 5 = 0101, 3 = 0011
print("Using a =", a, "and b =", b)
print("Binary of a:", format(a, '04b'))
print("Binary of b:", format(b, '04b'), "\n")

# Basic bitwise operations
print("a & b  ->", a & b,  "Binary:", format(a & b, '04b'))   # AND
print("a | b  ->", a | b,  "Binary:", format(a | b, '04b'))   # OR
print("a ^ b  ->", a ^ b,  "Binary:", format(a ^ b, '04b'))   # XOR
print("~a     ->", ~a,     "Binary:", format(~a & 0xff, '08b'))  # NOT

# Shifts
print("a << 1 ->", a << 1, "Binary:", format(a << 1, '04b'))
print("a >> 1 ->", a >> 1, "Binary:", format(a >> 1, '04b'))

# User input example
x = int(input("\nEnter first integer: "))
y = int(input("Enter second integer: "))

print("\nBitwise results:")
print(x, "&", y, "=", x & y)
print(x, "|", y, "=", x | y)
print(x, "^", y, "=", x ^ y)
print("~", x, "=", ~x)
print(x, "<< 1 =", x << 1)
print(x, ">> 1 =", x >> 1)

print("\nBitwise operators: &, |, ^, ~, <<, >>")
```

Identity Operators in Python

Identity operators are used to check whether two variables refer to the same object in memory, not just whether they hold the same value. In Python, everything is an object, and each object has a unique memory address (or identity).

There are two identity operators:

- is → Returns True if both variables point to the *same object*.
- is not → Returns True if both variables point to *different objects*.

It's important to note the difference between is and ==:

- == checks values are they equal in content?
- is checks identity are they stored in the same memory location?

For immutable types like integers and strings, Python sometimes reuses objects (a process called *interning*). For example, small integers (-5 to 256) or identical strings might share the same memory reference. Mutable types like lists, dictionaries, or sets, however, always create new objects even if they have the same content.

The is operator is also commonly used when comparing with None, since None is a singleton object in Python there's only one instance of it in memory.

Lesson Program: Identity Operators in Python.

Source Code:

identityOperators.py

```python
# Identity Operators in Python

print("\n===== IDENTITY OPERATORS IN PYTHON =====")

# Integers
a, b = 10, 10
print("a =", a, "b =", b)
print("a == b ->", a == b)
print("a is b ->", a is b)
print("a is not b ->", a is not b, "\n")

# Strings
x, y = "hello", "hello"
print("x =", x, "y =", y)
print("x == y ->", x == y)
print("x is y ->", x is y, "\n")

# Lists (mutable objects)
list1 = [1, 2, 3]
list2 = [1, 2, 3]
print("list1 == list2 ->", list1 == list2)
print("list1 is list2 ->", list1 is list2)
print("list1 is not list2 ->", list1 is not list2, "\n")

# None comparison
val = None
print("val is None ->", val is None)
print("val is not None ->", val is not None, "\n")

# Memory check
num1, num2, num3, num4 = 256, 256, 300, 300
print("num1 is num2 ->", num1 is num2)
print("num3 is num4 ->", num3 is num4)

print("\nUse 'is' for identity and '==' for value comparison.\n")
```

Membership Operators in Python

Membership operators are used to check whether a value exists inside a sequence such as a *string, list, tuple, set,* or *dictionary*.

Python gives two operators:

- in → True if the value is present
- not in → True if the value is *not* present

These checks work differently depending on the sequence:

- In strings, it checks for characters or substrings.
- In lists/tuples, it checks for exact elements.
- In sets, checks membership (sets are optimized for fast lookup).
- In dictionaries, membership only checks keys, not values.
  (So `"John" in student` is False unless `"John"` is a *key*, not a value.)

Useful when searching, filtering, validating user input, or quickly checking existence.

Lesson Program: Membership Operators in Python.

Source Code:

membershipOperators.py

```python
# ===== MEMBERSHIP OPERATORS IN PYTHON =====

# Membership operators help check if a value exists inside sequences
# Operators:  'in'  → True if value exists
#             'not in' → True if value does NOT exist

# ---- String Example ----
text = "python programming"
print("'p' in text:", 'p' in text)      # checks character
print("'java' not in text:", 'java' not in text)  # substring not found

# ---- List Example ----
fruits = ["apple", "banana", "cherry"]
print("'banana' in fruits:", 'banana' in fruits)  # element exists

# ---- Tuple Example ----
nums = (10, 20, 30)
print("20 in nums:", 20 in nums)  # tuple membership check

# ---- Dictionary Example ----
student = {"name": "John", "age": 23}
print("'name' in student:", 'name' in student)  # checks keys only

# ---- Set Example ----
colors = {"red", "green", "blue"}
print("'yellow' not in colors:", 'yellow' not in colors)  # fast lookup

# ---- User Input Example ----
word = input("Enter a word: ")
if 'a' in word:                    # checking character presence
    print("The letter 'a' is present.")
else:
    print("The letter 'a' is not present.")

print("\nUse 'in' or 'not in' to test membership in strings, lists, tuples, sets, and dictionary.")
```

Python Operators Summary

Operators are symbols that let Python perform actions on values, like calculations, comparisons, and logical checks.

- Arithmetic operators (+, –, *, /, //, %, **) handle basic math and are used whenever numbers are processed.
- Assignment operators (=, +=, -=, *=, etc.) update variable values without needing extra steps, making code cleaner and shorter.
- Comparison operators (==, !=, >, <, >=, <=) check how two values relate and always return True or False, helping in decisions and conditions.
- Logical operators (and, or, not) combine or reverse conditions, commonly used inside if statements or loops to create meaningful checks.
- Identity operators (is, is not) compare whether two variables point to the same object in memory, not just the same value.
- Membership operators (in, not in) check if a value exists inside sequences like strings, lists, sets, or dictionary keys.
- Bitwise operators (&, |, ^, ~, <<, >>) work at the binary level; less common but useful in specific tasks like flags or performance-heavy logic.
- Python also follows operator precedence, meaning some operations run before others (like ** before *, and * before +), and parentheses can be used to control evaluation.

Overall, operators help Python evaluate expressions, make decisions, update values, and interact with data efficiently across all kinds of programs.

Control Flow in Python

Control flow refers to the way Python decides which statements to execute and in what order. By default, a Python program runs from top to bottom, one line after another. But real programs need decisions, repetitions, and ways to skip or stop certain actions. Control flow makes this possible.

Python provides three major kinds of control flow mechanisms:

1. Conditional Statements

These allow the program to choose different paths based on conditions. Using if, elif, and else, Python can decide **what to execute** depending on whether a condition is True or False.
This is essential for decision-making, like checking passwords, validating input, or handling different cases in logic.

2. Looping Statements

Loops help repeat actions without rewriting code many times.
Python provides two loops:

- for → used when you know the number of repetitions (e.g., iterating through a list or range)
- while → used when you want to repeat until a condition becomes False

Loops are fundamental for tasks like counting, searching, automation, and processing collections of data.

3. Jump Statements

Python includes special statements to modify how loops behave:

- break → stops a loop immediately
- continue → skips the current iteration and moves to the next
- pass → does nothing; used as a placeholder

These help control loop behavior, especially when handling exceptions, filtering values, or creating placeholder structures during development.

Together, conditional statements, loops, and jump statements form the backbone of decision-making and repetition in Python programs. Control flow transforms code from simple linear execution into dynamic, intelligent behavior.

Lesson Program: Control Flow in Python.

Source Code:

controlFlow.py

```python
# Control Flow in Python

print("\n===== CONTROL FLOW IN PYTHON =====")

print("Control flow changes the order in which code executes.\n")

# Types of control flow
print("1. Conditional Statements → if, elif, else")
print("2. Looping Statements     → for, while")
print("3. Jump Statements        → break, continue, pass\n")

# Example values
temperature = 30
count = 3
weather = "sunny"

# Conditional example
if weather == "sunny":
    print("Bright day!")
elif weather == "rainy":
    print("Carry umbrella.")
else:
    print("Stay warm.")

# Looping examples
print("\nNumbers 1 to 5:")
for i in range(1, 6):
    print(i, end=" ")

print("\nCountdown:")
while count > 0:
    print(count, end=" ")
    count -= 1

# Jump statements
print("\n\nJump statements demo:")
for i in range(1, 6):
    if i == 2:
        continue
    if i == 4:
        break
    print("Number:", i)

# pass example
for j in range(3):
    pass

print("\nControl flow helps Python make decisions and repeat tasks.\n")
```

Conditionals in Python

Conditional statements allow a program to **make decisions**. They check whether a condition is True or False, and based on that, choose which block of code should run.

if Statement

The simplest form is the **if statement**, which runs a block **only when the condition is True**.

The structure looks like this:

```
if condition:
    # code to execute when condition is True
```

The condition is usually an expression using comparison or logical operators. Python uses indentation to define what code belongs inside the if block. If the condition evaluates to False, Python simply skips the block and continues with the rest of the program.

IF statements are used everywhere, checking login details, validating input, showing different messages, or making choices based on data.

Lesson Program: Conditionals if in Python.

Source Code:

ifStatement.py

```python
# Conditional Statements in Python - IF

print("\n===== IF STATEMENT IN PYTHON =====")

# Example 1: basic IF
age = int(input("Enter your age: "))
if age >= 18:
    print("Eligible to vote.")

# Example 2: temperature check
temp = float(input("\nEnter temperature: "))
if temp > 30:
    print("It's hot today.")

# Example 3: boolean condition
is_student = True
if is_student:
    print("\nStudent discount available.")

# Example 4: independent IFs
marks = int(input("\nEnter marks: "))
if marks >= 90: print("Grade A+")
if marks >= 75: print("Grade A")
if marks >= 60: print("Grade B")
if marks >= 40: print("Grade C")
if marks < 40: print("Fail")

# Example 5: even/odd
num = int(input("\nEnter a number: "))
if num % 2 == 0: print("Even number")
if num % 2 != 0: print("Odd number")

print("\nIF executes only when condition is True.\n")
```

if-else Statement

The if-else statement is used when the program must choose between two possible actions. If the condition is True, Python runs the if block; otherwise, it runs the else block. This creates a two-way decision, allowing the program to respond differently depending on a situation.

The general structure is:

```
if condition:
    # code when condition is True
else:
    # code when condition is False
```

IF–ELSE is commonly used for checking user input, validating age or passwords, comparing numbers, detecting even/odd numbers, and handling basic decision-making. Only one of the two blocks executes, never both.

Indentation is what tells Python which statements belong to the if block and which belong to the else block.

Lesson Program: Conditionals if-else in Python.

Source Code:

ifElseStatement.py

```python
# IF-ELSE Statement in Python

print("\n===== IF-ELSE STATEMENT IN PYTHON =====")

# Even or odd
n = int(input("Enter a number: "))
if n % 2 == 0:
    print("Even")
else:
    print("Odd")

# Voting eligibility
age = int(input("\nEnter age: "))
if age >= 18:
    print("Eligible")
else:
    print("Not eligible")

# Compare numbers
a = int(input("\nEnter first number: "))
b = int(input("Enter second number: "))
if a > b:
    print("First is greater")
else:
    print("Second is greater or equal")

print("\nIF-ELSE executes only one block.\n")
```

elid Ladder Statement

The if-elif-else ladder is used when a program must choose one option out of many possible conditions. Instead of writing several separate if statements, the ladder checks conditions from top to bottom, and the first True condition decides which block runs.

The general structure is:

```
if condition1:
   # runs if condition1 is True
elif condition2:
   # runs if condition1 is False and condition2 is True
elif condition3:
   # runs if above conditions are False and condition3 is True
else:
   # runs only if all conditions are False
```

Only one block executes, whichever condition is met first. This structure is useful for cases like grading systems, menu choices, category checks, and value ranges.

Lesson Program: Conditionals elif Ladder in Python.

Source Code:

elifStatement.py

```python
# IF-ELIF-ELSE Ladder in Python

print("\n===== IF-ELIF-ELSE LADDER =====")

# Grading example
marks = int(input("Enter marks: "))

if marks >= 90:
    print("Grade A+")
elif marks >= 75:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
elif marks >= 40:
    print("Grade C")
else:
    print("Fail")

# Temperature example
temp = float(input("\nEnter temperature: "))

if temp > 35:
    print("Too hot")
elif temp > 25:
    print("Warm")
elif temp > 15:
    print("Pleasant")
elif temp > 5:
    print("Cold")
else:
    print("Freezing")

print("\nOnly one block runs based on the first True condition.\n")
```

Nested if-else Statement

A nested if-else statement means placing one **if or else block inside another**. It is used when decisions depend on more than one related condition. The outer condition must be True before Python even checks the inner one.

The general structure is:

```
if condition1:
    if condition2:
        # inner block
    else:
        # inner else block
else:
    # outer else block
```

Nested conditions help create multi-level decision logic, such as checking:

- pass/fail first, then checking grade
- login first, then checking user type
- validating main condition before detailed checks

Indentation is crucial because it shows Python which statement belongs to which level of decision.

Lesson Program: Conditionals Nested if-else in Python.

Source Code:

nestedIfElseStatement.py

```python
# Nested IF-ELSE in Python

print("\n===== NESTED IF-ELSE =====")

# Grading example
marks = int(input("Enter marks: "))

if marks >= 40:
    print("Pass")
    if marks >= 90:
        print("Grade A+")
    elif marks >= 75:
        print("Grade A")
    elif marks >= 60:
        print("Grade B")
    else:
        print("Grade C")
else:
    print("Fail")

# Temperature example
temp = float(input("\nEnter temperature: "))

if temp >= 0:
    print("Above freezing")
    if temp > 35:
        print("Very hot")
    else:
        print("Normal")
else:
    print("Freezing")

print("\nNested IF checks inner conditions only if the outer condition is True.\n")
```

Short Hand Conditionals Statements

Python allows writing simple conditional statements in a single line. These are useful when the action is short and does not need multiple lines of code.

There are three main forms:

1. Single-line IF

Used when only one statement needs to run if the condition is True.

```
if condition: statement
```

2. Short-hand IF-ELSE (Ternary Expression)

Chooses between two values in a single line.

```
value_if_true if condition else value_if_false
```

3. Nested Ternary Expression

Used to check multiple conditions in a compact form.

```
value1 if condition1 else value2 if condition2 else value3
```

These expressions make code shorter and are best used for simple decisions, quick checks, or assigning values based on conditions.

Lesson Program: Conditionals Short Hand if-else in Python.

Source Code:

shortHandIfStatement.py

```python
# Short-hand IF / Conditional Expressions

print("\n===== SHORT-HAND IF / TERNARY =====")

# Single-line IF
x = 10
if x > 5: print("x > 5")

# Short-hand IF-ELSE (ternary)
a, b = 7, 12
max_val = a if a > b else b
print("Max value:", max_val)

# Nested ternary
num = int(input("\nEnter a number: "))
result = "Positive" if num > 0 else "Negative" if num < 0 else "Zero"
print("The number is:", result)

print("\nShort-hand IF is best for simple, single-line decisions.\n")
```

Python Conditionals Summary

- Conditional statements help a program make decisions based on conditions.
- Python checks conditions using Boolean results (True / False) and runs the appropriate block of code.
- if statement runs a block only when the condition is True.
- if-else provides two-way decision making, one block runs if the condition is True, the other runs when it is False.
- if-elif-else ladder is used when there are multiple possible conditions, checked from top to bottom. Only the first True block runs.
- Nested if statements place one if/else inside another, used for checking multiple related conditions in a structured hierarchy.
- Short-hand if lets you write a simple one-line if statement.
- Ternary (short-hand if-else) provides quick value selection in one line:
  `result = value_if_true if condition else value_if_false`
- Nested ternary expressions allow multiple conditions in a compact form but should be used carefully to avoid unreadable code.
- Indentation is essential, it defines which lines belong to each conditional block.
- Conditions often use relational operators (>, <, ==, etc.) and logical operators (and, or, not) to form more complex checks.

In short, conditionals form the foundation of decision-making in Python, allowing your program to react differently depending on values, comparisons, and user input.

Loops in Python

Loops allow a program to execute a block of code repeatedly. Instead of writing the same statements again and again, loops automate repetition.

Python provides two main looping constructs:

Why Loops?

Loops are used when a task needs to be repeated multiple times, such as:

- printing a message several times
- iterating over a list of items
- performing calculations for many elements
- reading characters from a string

Two Types of Loops in Python

Loop Type	Purpose
for loop	Used when the number of repetitions is known or when iterating over a sequence (list, string, tuple, range, etc.)
while loop	Used when repetition depends on a condition and may run any number of times

The for Loop

The for loop iterates over a sequence of elements. Python does not use index-based looping by default like other languages, instead, it directly takes each element one by one.

Basic structure:

```
for variable in sequence:
    # code block
```

Understanding range()

range() generates a sequence of numbers for the loop:

```
range(start, stop, step)
```

- start → Beginning number (default 0)
- stop → End number (excluded)
- step → Increment/decrement (default 1)

Lesson Program: For Loop in Python.

Source Code:

forLoop.py

```python
# For Loop in Python

print("\n===== FOR LOOP IN PYTHON =====")

# range()
for i in range(3):
    print(i, end=" ")

# list
print("\n")
for fruit in ["Apple", "Banana"]:
    print(fruit, end=" ")

# Print 1 to 5
for i in range(1, 6):
    print(i, end=" ")

# Print characters of a string
print("\n")
for ch in "PYTHON":
    print(ch, end=" ")

# Print a multiplication table
n = int(input("\n\nEnter a number: "))
for i in range(1, 11):
    print(n, "x", i, "=", n * i)

print("\nFor loop used to iterate over sequences and ranges.\n")
```

The While Loop

A while loop is used when the number of repetitions is not known in advance. It keeps executing a block of code as long as a condition remains True.

Basic structure:

```
while condition:
    # code block
```

How it works

- The condition is checked first.
- If it is True → the loop runs.
- After executing the block, the condition is checked again.
- When the condition becomes False → the loop stops.

Key Points

- We must update the condition variable inside the loop, otherwise an infinite loop occurs.
- while is useful in situations where repetition depends on a condition, such as:
    - waiting for correct input
    - countdown timers
    - reading data until a limit is reached
    - loops that depend on user choices

Infinite Loop Example

```
while True:
    print("Runs forever")   # Must break using `break`
```

The loop must eventually stop using logic or a break statement.

Lesson Program: While Loop in Python.

Source Code:

whileLoop.py

```python
# WHILE Loop in Python

print("\n===== WHILE LOOP =====")

# Counting 1 to 5
i = 1
while i <= 5:
    print(i, end=" ")
    i += 1

# Countdown
print("\n")
x = 5
while x > 0:
    print(x, end=" ")
    x -= 1

# Repeat until correct password
print("\n")
pwd = ""
while pwd != "python": pwd = input("Enter password: ")
print("Access Granted")

# Sum of N natural numbers
print()
n = int(input("Enter a number: "))
sum_n, j = 0, 1
while j <= n:
    sum_n += j
    j += 1
print("Sum:", sum_n)

print("\nWhile loop repeats as long as the condition remains True.\n")
```

Nested Loops in Python

A nested loop means one loop placed inside another loop. The outer loop controls how many times a major action repeats, and for each iteration of the outer loop, the inner loop runs completely.

How It Works

```
for i in range(...):     ← Outer loop
  for j in range(...): ← Inner loop
    # code block
```

- The inner loop executes all its iterations for each single run of the outer loop.
- After the inner loop finishes, the outer loop moves to its next iteration.

Key Uses of Nested Loops

- Pattern printing
- Working with tables or grids
- Traversing 2D lists (matrices)
- Generating combinations, e.g., sizes & colors
- Repeated actions on grouped data

Note

Nested loops increase execution time because the inner loop runs multiple times for each outer loop iteration. Use them carefully in large data situations.

Lesson Program: Nested Loop in Python.

Source Code:

nestedLoop.py

```python
# Nested Loops in Python

print("\n===== NESTED LOOPS =====")

# For-for nested loop
for i in range(1, 3):
    for j in range(1, 4):
        print(i, j, end=" ")
    print()

# Nested loop for 2D list
matrix = [[1, 2], [3, 4]]
for row in matrix:
    for val in row:
        print(val, end=" ")
    print()

# Generating combinations
colors = ["Red", "Blue"]
sizes = ["S", "M"]
for c in colors:
    for s in sizes:
        print(c, s)

print("\nInner loop runs fully for each outer loop iteration.\n")
```

Jump Statements in Python

Jump statements are used to change the normal flow of loops. Instead of running from the first iteration to the last, these statements let us:

- exit a loop immediately
- skip specific iterations
- keep a placeholder for future code

1. break

- Stops the loop completely at the moment it is executed.
- Used when a required value is found or when we need to stop a loop early.

2. continue

- Skips the current iteration and moves to the next one.
- Used when we want to ignore specific cases without stopping the entire loop.

3. pass

- Does nothing. It is just a placeholder.
- Useful in empty loops, future code blocks, or when logic will be added later.

These statements work with both for and while loops and help manage control flow efficiently.

Lesson Program: Jump Statements in Python.

Source Code:

jumpStatements.py

```python
# Jump Statements: break, continue, pass

print("\n===== JUMP STATEMENTS =====")

# break example
for i in range(1, 10):
    if i == 5:
        break
    print(i, end=" ")

# continue example
print("\n")
for i in range(1, 10):
    if i % 2 == 0:
        continue
    print(i, end=" ")

# pass example
print("\n")
for ch in "Hi":
    if ch == "i":
        pass
    print(ch, end=" ")

print("\n\nbreak stops, continue skips, pass does nothing.\n")
```

Loop Else in Python

In Python, both for and while loops can have an else block. The else part runs only when the loop completes normally (i.e., without a break statement).

How it works

```
for item in sequence:
    # loop body
else:
    # runs only if loop was not stopped using break
```

Same for while loops.

Key Ideas

- The else block is not tied to an if inside the loop.
- It executes only if the loop ends naturally, not because of break.
- Common use case: searching — if something is not found in the loop, else can show a "not found" message.

continue does not stop the loop, so else still runs.

Lesson Program: Loop Else in Python.

Source Code:

loopElse.py

```python
# Loop Else in Python

print("\n===== LOOP ELSE =====")

# for loop search example
nums = [2, 4, 6, 8]
for n in nums:
    if n == 5:
        print("Found 5")
        break
else:
    print("5 not found in list")

# while loop example
print()
x = 3
while x > 0:
    print(x, end=" ")
    x -= 1
else:
    print("\nCountdown finished")

for user in users:
    if user == target:
        print("User found")
        break
else:
    print("User not found")

print("\nElse runs only when loop completes without break.\n")
```

Loops Summary

- Loops repeat a block of code multiple times; useful for automation and iteration.
- for loop iterates over sequences like lists, strings, tuples, sets, dictionaries, or numbers generated using `range()`.
- while loop repeats as long as a condition stays True; used when the number of repetitions is unknown in advance.
- Both loops require proper indentation and often need updating variables to avoid infinite loops.
- Nested loops are loops inside loops; the inner loop runs completely for each iteration of the outer loop, useful for patterns, matrices, and combinations.
- Jump Statements:
    - `break` stops the loop immediately.
    - `continue` skips the current iteration and moves to the next.
    - `pass` does nothing, used as a placeholder.
- else with loops executes only if the loop finishes normally without a `break` (often used in search operations).
- `range(start, stop, step)` is commonly used with for loops to control where counting starts, ends, and how much it increments or decrements.
- Loops should be used efficiently; avoid unnecessary nested loops or conditions that may affect performance.

Data Structures in Python

A data structure is a way of organizing, storing, and managing data so that it can be used efficiently. Different data structures allow faster searching, insertion, deletion, updating, or arrangement of data depending on the requirement.

Why Data Structures?

- To store data efficiently
- To access and modify data quickly
- To manage large amounts of information
- To perform operations like searching, sorting, grouping, linking

Types of Data Structures in Python

Python provides two categories of data structures:

1) Built-in Data Structures

These are provided by Python and are ready to use.

Data Structure	Ordered?	Mutable?	Stores
String	Yes	No	Characters
List	Yes	Yes	Collection of elements
Tuple	Yes	No	Collection of elements
Set	No	Yes	Unique elements
Dictionary	No	Yes	Key–value pairs

2) User-Defined Data Structures

These are created by programmers using Python's classes and objects.

Examples include:

- Stack
- Queue
- Linked List
- Tree
- Graph
- Hash Table

These structures help solve complex problems and implement algorithms according to specific needs.

Key Concepts

- Ordered vs Unordered: Controls how items are stored internally.
- Mutable vs Immutable: Defines whether the data can be changed after creation.
- Uniqueness: Some structures like sets do not allow duplicates.
- Key-value mapping in dictionaries allows fast access using keys.

Lesson Program: Data Structures in Python.

Source Code:

dataStructures.py

```python
# Data Structures in Python

print("\n===== DATA STRUCTURES =====")

# Built-in Examples
s = "Python"                    # String
l = [10, 20, 30]                # List
t = (1, 2, 3)                   # Tuple
st = {10, 20, 20, 30}           # Set (duplicates removed)
d = {"name": "John", "age": 23}  # Dictionary

print(s)
print(l)
print(t)
print(st)
print(d)

# User-defined Data Structure: Stack (Using Class)
class Stack:  1 usage
    def __init__(self):
        self.items = []
    def push(self, val):  2 usages
        self.items.append(val)
    def pop(self):  2 usages (1 dynamic)
        return self.items.pop()

# Using the Stack
stk = Stack()
stk.push(100)
stk.push(200)
print("\nStack pop:", stk.pop())  # Last in, first out
```

Strings in Python

A string in Python is a sequence of characters enclosed in quotes (' ', " ", or """ """). It can contain letters, digits, symbols, whitespace, or even be empty. Strings are immutable, meaning once created, their characters cannot be directly modified.

Python supports:

Single quotes 'Hello'
Double quotes "Hello"
Triple quotes """Hello""" for multi-line text

Each character in a string has an index:

- Starts at 0 for the first character
- Supports negative indexes -1 for last character

Python allows slicing strings to extract parts: text[start:end:step].

Strings come with many built-in methods for text processing:

- lower(), upper() change case
- strip() removes extra spaces
- find() searches text
- replace() changes text
- split() breaks into a list
- count() counts occurrences

To format dynamic text, Python supports:

- f-strings (modern and recommended)
- .format() method (older but useful)

Overall, strings are one of Python's most common data types used for names, messages, file paths, logs, text processing, and more.

Lesson Program: Strings in Python.

Source Code:

strings.py

```python
# Strings in Python

# Creating strings
s1 = "Python"
msg = """Learning
Strings in Python"""

print(s1)
print(msg)

# Indexing & slicing
print(s1[0], s1[-1])           # first & last char
print(s1[1:4], s1[::-1])       # slice & reverse

# Useful methods
t = "  Hello Python  "
print(t.strip(), t.lower(), t.replace( _old: "Python",  _new: "Java"))

# Formatting
name, age = "John", 23
print(f"{name} is {age} years old")           # f-string
print("Name: {}, Age: {}".format( *args: name, age))  # format()
```

Lists in Python

A list is one of the most flexible and widely used data structures in Python. It is an ordered, mutable collection that can store multiple items of any data type. Because lists can grow, shrink, and be modified easily, they are used in almost every Python program.

Key Features of Lists

- Ordered → items maintain a fixed index (0, 1, 2, ...)
- Mutable → items can be changed after creation
- Allow duplicates → same value can appear multiple times
- Can store mixed types → integers, strings, floats, booleans, other lists, etc.
- Dynamic in size → you can add or remove items anytime

Creating Lists

Lists are written inside square brackets:

```
numbers = [10, 20, 30]
mixed = [10, "Hello", 3.14, True]
empty = []
nested = [[1, 2], [3, 4]]
```

Accessing List Elements

Python supports:

- Indexing: list[0], list[-1]
- Slicing: list[1:4], list[::-1], etc.

Useful List Methods

Lists support many built-in operations:

- append(), insert() → add items
- remove(), pop() → delete items
- sort(), reverse() → reorder items
- count(), index() → search operations
- extend() → combine lists

Looping Through Lists

Lists can be iterated using:

- for item in list
- for i in range(len(list))

Nested Lists

Lists can store other lists, forming a 2D structure (matrix-like):

matrix = [[1,2,3], [4,5,6], [7,8,9]]

Access with two indexes:

matrix[1][2] → 6

Lists are foundational for data handling, algorithms, and real-world applications like storing records, processing user input, building menus, handling dynamic collections, and more.

Lesson Program: Lists in Python.

Source Code:

lists.py

```python
# Lists in Python

# Creating lists
nums = [10, 20, 30]
mixed = [10, "Python", 3.14]
nested = [[1, 2], [3, 4]]

print(nums)
print(mixed)
print(nested)

# Indexing & slicing
print(nums[0], nums[-1])
print(nums[1:3], nums[::-1])

# List methods
fruits = ["apple", "banana"]
fruits.append("mango")
fruits.insert( _index: 1,  _object: "grapes")
fruits.remove("banana")
print(fruits)

# Iterating
for f in fruits:
    print(f)

# Nested list access
matrix = [[1, 2], [3, 4]]
print(matrix[1][0])
```

Tuples in Python

A tuple is an ordered collection of items, similar to a list, but with one key difference: tuples are immutable, meaning their elements cannot be changed after creation. Because of this, tuples are safer, faster, and often used for *fixed* collections of data.

Key Features of Tuples

- Ordered → items maintain their positions
- Immutable → values cannot be changed, added, or removed
- Allow duplicates
- Can store mixed data types
- Faster than lists due to immutability
- Hashable (if containing only immutable items), so they can be used as dictionary keys

Creating Tuples

Tuples use parentheses ():

```
t = (10, 20, 30)
mixed = (10, "Python", 3.5)
```

A single-element tuple must include a comma:

```
x = (5,)      # tuple
y = (5)       # not a tuple, just an integer
```

Accessing Tuple Elements

- Indexing: t[0], t[-1]
- Slicing: t[1:3], t[::-1]

Immutability

You cannot:

- change a value
- add elements
- remove elements

But you *can* create a new tuple by combining others:

```
t = (1, 2, 3)
new_t = t + (4, 5)
```

Useful Tuple Methods

Tuples support only two built-in methods:

- count(value) → number of occurrences
- index(value) → position of first occurrence

When to Use Tuples

- When data must not change (constants, configurations)
- When using as keys in dictionaries
- When returning multiple values from functions

Tuples are simple, lightweight, and perfect for fixed, structured data.

Lesson Program: Tuples in Python.

Source Code:

tuples.py

```python
# Tuples in Python

# Creating tuples
nums = (10, 20, 30)
mixed = (10, "Python", 3.14)
single = (5,)
nested = (1, 2, (3, 4))

print(nums)
print(mixed)
print(single)
print(nested)

# Accessing elements
fruits = ("apple", "banana", "mango")
print(fruits[0], fruits[-1])
print(fruits[1:3], fruits[::-1])

# Immutability example (create new tuple)
t = (1, 2, 3)
new_t = t + (4, 5)
print(new_t)

# Tuple methods
vals = (10, 20, 10, 30)
print(vals.count(10))
print(vals.index(30))
```

Sets in Python

A set is a collection designed for speed and uniqueness. Unlike lists and tuples, sets don't care about order or indexing. They simply hold items, making sure no duplicates sneak in.

Key Characteristics of Sets

- Unordered → items don't have fixed positions
- Mutable → you can add or remove items
- No duplicates → automatic cleanup of repeated values
- Fast membership checks → `in` is extremely efficient

Sets use curly braces `{}`, except for empty sets where we use `set()`.

Creating Sets

```
s = {1, 2, 3}
empty = set()
mixed = {10, "Python", 3.14}
```

Duplicates disappear automatically:

{1, 1, 2, 2} → {1, 2}

Modifying a Set

- `add(x)` → adds element
- `remove(x)` → removes (errors if missing)
- `discard(x)` → removes safely
- `pop()` → removes a random item
- `clear()` → empties the set

Set Operations (Very Important)

Sets support rich mathematical operations:

- Union → A ∪ B → all items
- Intersection → A ∩ B → common items
- Difference → A - B → items only in A
- Symmetric difference → items in either set, not both

Relationship Methods

- `issubset()`
- `issuperset()`
- `isdisjoint()`

These help compare sets easily.

When to Use Sets

- Removing duplicates
- Fast lookups
- Mathematical operations

- Checking relationships between data groups

Sets keep things clean, fast, and unique.

Lesson Program: Sets in Python.

Source Code:

sets.py

```python
# Sets in Python

# Creating sets
nums = {10, 20, 30}
mixed = {10, "Python", 3.14}
empty = set()
dup = {1, 2, 2, 3}

print(nums, mixed, empty, dup)

# Adding & removing
fruits = {"apple", "banana", "mango"}
fruits.add("orange")
fruits.remove("banana")
fruits.discard("grapes")   # safe
print(fruits)

item = fruits.pop()
print("Popped:", item)

# Set operations
A = {1, 2, 3, 4}
B = {3, 4, 5}

print(A.union(B))
print(A.intersection(B))
print(A.difference(B))
print(A.symmetric_difference(B))

# Relationship methods
print(A.issubset(B))
print(A.issuperset(B))
print(A.isdisjoint(B))

# update() and copy()
A.update(B)
print(A)
copyA = A.copy()
print(copyA)
```

Dictionaries in Python

A dictionary is Python's powerhouse data structure for storing *key–value* pairs. Instead of accessing data by positions (like lists), dictionaries let you access values quickly using meaningful *keys*. This makes them perfect for real-world structured data such as user profiles, product info, marksheets, configs, and more.

Key Characteristics

- Unordered (Python 3.7+ preserves insertion order, but conceptually unordered)
- Mutable (you can add, update, delete entries)
- Keys must be unique
- Values can be of any data type
- Defined using curly braces:
  {key: value, key: value}

Keys vs Values

- Keys are usually strings or numbers (must be immutable)
- Values can be anything: numbers, lists, dictionaries, even functions

Creating Dictionaries

```
student = {"name": "Aditya", "age": 23}
profile = dict(username="john", active=True)
empty = {}
```

Nested dictionaries allow multi-level structured data.

Accessing Data

- dict[key] for direct access
- dict.get(key) for safe access (doesn't throw error)
- keys(), values(), items() expose all components

Adding / Updating

- dict[key] = value
- update({...}) for multiple changes
- setdefault() for adding only if key missing

Removing

- pop(key) → remove and return value
- popitem() → removes last inserted pair
- del dict[key]
- clear() → empty the dictionary

Iteration

Loop through:

- keys
- values
- both (using `.items()`)

With enumerate(), you can add indexing too.

Nested Dictionaries

Dictionaries inside dictionaries help represent structured data (like JSON).

Common Uses

- Representing objects
- Config files
- Word-frequency counters
- JSON data handling

Dictionaries are fast, expressive, and ideal for labelled data.

Lesson Program: Dictionaries in Python.

Source Code:

dictionaries.py

```python
# Dictionaries in Python

# Creating dictionaries
student = {"name": "Aditya", "age": 23, "course": "CS"}
person = dict(name="Sneha", age=21)
empty = {}
school = {"s1": {"name": "Asha", "marks": 82}}

print(student, person, empty, school)

# Accessing values
print(student["name"])
print(student.get("course"))
print(student.get("phone", "Not Provided"))

print(list(student.keys()))
print(list(student.values()))
print(list(student.items()))

# Adding & updating
student["phone"] = "98765"
student["age"] = 24
student.update({"city": "Pune"})
student.setdefault("dept", "Engineering")
print(student)

# Removing elements
student.pop("phone", None)
student.popitem()
if "city" in student:
    del student["city"]
print(student)

# Iterating
emp = {"id": 101, "name": "Rina", "role": "Engineer"}
for k in emp: print(k)
for v in emp.values(): print(v)
for k, v in emp.items(): print(k, v)

# Nested dictionary access
company = {"d1": {"manager": "Amit"}}
print(company["d1"]["manager"])

# Frequency counter (short)
text = input("Enter text: ").split()
freq = {}
for w in text:
    freq[w] = freq.get(w, 0) + 1
print(freq)
```

Data Structures Summary

Data structures in Python define how data is stored, organized, and accessed efficiently. They help manage collections of values and support different operations like searching, updating, and iteration.

• Python provides built-in data structures that are ready to use:
• String (str) – Ordered, immutable sequence of characters. Supports indexing, slicing, and many useful methods for text processing.
• List – Ordered and mutable. Allows adding, removing, and modifying items. Perfect for dynamic collections.
• Tuple – Ordered and immutable. Useful for fixed data that should not change. Supports indexing and slicing.
• Set – Unordered, mutable collection of unique elements. Supports mathematical operations like union and intersection.
• Dictionary (dict) – Unordered mapping of key-value pairs. Keys are unique; values can be any type. Ideal for structured and labelled data.
• Mutability distinguishes these structures:
• Mutable: list, set, dictionary
• Immutable: string, tuple
• Indexing and slicing work for ordered structures (string, list, tuple) but not for sets or dictionaries.
• Collections support useful operations such as iteration, membership checking, updating, copying, and combining.
• Python also supports user-defined data structures like stacks, queues, linked lists, and trees, usually built using lists, classes, or modules like collections.
• Choosing the correct data structure improves performance, readability, and the efficiency of algorithms.

In short, Python's data structures give you flexible and powerful tools to organize data, whether you need order, uniqueness, immutability, key-based access, or dynamic resizing.

Functions in Python

A function in Python is a named block of code that performs a specific task. Instead of writing the same set of instructions repeatedly, you can place them inside a function and reuse them whenever needed. This helps keep programs shorter, cleaner, and easier to understand.

Functions improve:

- Reusability – write once, use many times
- Readability – logic is grouped with meaningful names
- Maintainability – changes are made in one place
- Modularity – large programs are divided into smaller parts

Functions are defined using the `def` keyword, followed by the function name and parentheses. A function only runs when it is called.

Function Parameters and Arguments

Functions can receive input values called parameters. When calling a function, the values passed are called arguments.

Python supports:

- Positional arguments – values passed in order
- Keyword arguments – values passed using parameter names
- Default arguments – parameters with predefined values

These features make function calls flexible and readable.

Variable-Length Arguments

Sometimes the number of inputs is not fixed. Python allows this using:

- `*args` – collects multiple positional arguments into a tuple
- `**kwargs` – collects multiple keyword arguments into a dictionary

This is useful when designing functions that need to handle varying data.

Return Statement

A function can send a result back to the caller using the `return` statement. Once `return` is executed, the function stops running.

If no return statement is used, Python automatically returns `None`. Returned values can be stored in variables, printed, or used in expressions.

Scope of Variables

Variables have a scope, which determines where they can be accessed.

- Local variables are created inside a function and exist only there
- Global variables are created outside functions and accessible everywhere

To modify a global variable inside a function, the `global` keyword is used.

Built-in Functions

Python provides many built-in functions to perform common tasks, such as:

- `len()` – length of a collection
- `sum()` – total of values
- `max()` / `min()` – largest or smallest value
- `sorted()` – returns sorted data
- `range()` – generates number sequences

These reduce the need to write common logic manually.

Functions with Loops

Functions can be used together with loops to perform repeated operations efficiently. This combination is very common in real-world programs such as calculations, reports, validations, and data processing.

In summary, functions are the backbone of structured programming in Python. They help organize logic, reduce duplication, and make programs scalable and easier to manage.

Lesson Program: Functions in Python.

Source Code:

functions.py

```python
# Functions in Python

def greet():  1 usage
    print("Hello from a function")

def add(a, b):
    return a + b

def info(name, course="Python"):  2 usages
    print(name, course)

def total(*nums):
    return sum(nums)

def details(**data):  1 usage
    print(data)

x = 10
def show():  1 usage
    x = 5
    print(x)

y = 20
def change():  1 usage
    global y
    y = 50

greet()
print(add( a: 3,  b: 4))
info("John")
info( name: "Asha",  course: "Java")
print(total( *nums: 10, 20, 30))
details(city="Mumbai", age=23)

show()
print(x)

change()
print(y)
```

Types of Functions & Lambda Functions in Python

Types of Functions in Python

Functions in Python can be broadly classified into the following types:

1. Built-in Functions
   These are functions already provided by Python and ready to use, such as print(), len(), sum(), max(), min(), range(), etc. They help perform common tasks quickly without writing extra code.
2. User-defined Functions
   These are functions created by the programmer using the def keyword. They allow customization of logic according to program needs and improve reusability.
3. Anonymous (Lambda) Functions
   These are small, unnamed functions created using the lambda keyword. They are used when a function is simple and required for a short duration.

Lambda (Anonymous) Functions

A lambda function is a one-line function without a name. It can take any number of arguments but contains only one expression.

Syntax:

```
lambda arguments : expression
```

Key points:

- No def keyword
- No function name
- Automatically returns the result
- Best for short, simple operations
- Commonly used with map(), filter(), and reduce()

map(), filter(), and reduce()

These are function-based tools used to process collections like lists and tuples.

• map() : Applies a function to every element of an iterable and returns a new iterable.

• filter() : Selects elements from an iterable for which a condition is True.

• reduce() : Combines all elements of an iterable into a single value by repeatedly applying a function.

(reduce() is available in the functools module.)

These tools make programs:

- Shorter
- Cleaner
- More functional in style

They are widely used in data processing and transformations.

Lesson Program: Advance Functions in Python.

Source Code:

advanceFunctions.py

```python
# Types of Functions & Lambda

# Lambda functions
square = lambda x: x * x
add = lambda a, b: a + b

print(square(4))
print(add( a: 5,  b: 6))

# map()
nums = [1, 2, 3, 4, 5]
doubled = list(map(lambda x: x * 2, nums))
print(doubled)

# filter()
evens = list(filter(lambda x: x % 2 == 0, nums))
print(evens)

# reduce()
from functools import reduce
total = reduce(lambda a, b: a + b, nums)
print(total)

# Combined example
scores = [45, 67, 89, 32]
bonus = list(map(lambda x: x + 5, scores))
passed = list(filter(lambda x: x >= 50, bonus))
final_total = reduce(lambda a, b: a + b, passed)

print(bonus)
print(passed)
print(final_total)
```

Recursion in Python

Recursion is a programming technique where a function calls itself to solve a problem. It works by breaking a large problem into smaller versions of the same problem until a simple condition is reached.

Every recursive function has two essential parts:

1. Base Case
    - The condition where recursion stops
    - Prevents infinite calls
    - Without a base case, recursion will cause an error
2. Recursive Case
    - The function calls itself with a smaller or simpler input
    - Moves the problem closer to the base case

How Recursion Works

- Each function call is placed on the call stack
- Python executes calls one by one
- When the base case is reached, values are returned backwards
- Final result is built while returning from calls

When to Use Recursion

Recursion is useful when:

- The problem is naturally repetitive
- The solution depends on solving smaller subproblems
- Data has a hierarchical structure

Common use cases:

- Mathematical problems (factorial, Fibonacci)
- File and directory traversal
- Searching and sorting algorithms
- Tree and graph traversal

Important Points for Beginners

- Always define a clear base case
- Ensure input moves toward the base case
- Recursion can be slower than loops due to function call overhead
- Deep recursion may cause stack overflow

In short, recursion provides elegant and readable solutions for problems that repeat in a structured way.

Lesson Program: Recursion Functions in Python.

Source Code:

recursionFunctions.py

```python
# Recursion in Python

# Countdown
def countdown(n):  2 usages
    if n == 0:
        return
    print(n)
    countdown(n - 1)

countdown(5)

# Sum of numbers
def sum_n(n):  2 usages
    if n == 0:
        return 0
    return n + sum_n(n - 1)

print(sum_n(5))

# Factorial
def factorial(n):  2 usages
    if n == 1:
        return 1
    return n * factorial(n - 1)

print(factorial(5))

# String character count
def count_char(text, ch, i=0):  2 usages
    if i == len(text):
        return 0
    return (1 if text[i] == ch else 0) + count_char(text, ch, i + 1)

print(count_char( text: "recursion",  ch: "r"))

# Number to binary
def to_binary(n):  2 usages
    if n == 0:
        return ""
    return to_binary(n // 2) + str(n % 2)

print(to_binary(13))
```

Functions Summary

- A function is a reusable block of code that performs a specific task. Functions help reduce repetition, improve readability, and make programs easier to maintain.
- Functions are defined using the `def` keyword and executed by calling the function name.
- Functions can accept parameters to receive input values and can return results using the `return` statement. If no value is returned, the function returns `None`.
- Python supports different ways of passing arguments:
    - Positional arguments (based on order)
    - Keyword arguments (using parameter names)
    - Default arguments (predefined values)
- Functions can accept a variable number of inputs using:
    - `*args` for multiple positional values
    - `**kwargs` for multiple key-value pairs
- Variables inside a function are local, while variables defined outside are `global`. The global keyword allows modification of global variables inside a function.
- Python provides many built-in functions such as `len()`, `sum()`, `max()`, `min()`, `sorted()`, and `range()` that simplify common tasks.
- Lambda functions are small, one-line anonymous functions used for simple operations, often with tools like `map()` and `filter()`.
- Recursion is a technique where a function calls itself to solve a problem using a base case and a recursive case.

In short, functions allow you to organize logic, reuse code, and write cleaner and more structured Python programs.